

# Scalable Kernelized One-Class LPP-NFST via Anchor-based Soft Grouping

## 1 Null Foley-Sammon Transform

Null Foley-Sammon Transform (NFST) is a powerful discriminative approach, particularly effective for the Small Sample Size (SSS) problem. NFST seeks a subspace where the within-class scatter vanishes, while the between-class scatter remains positive.

### 1.1 Scatter Matrices

Consider a training set  $\mathbf{X} = [\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n] \in \mathbb{R}^{d \times n}$  belonging to  $c$  classes  $\{C_1, \dots, C_c\}$ . Let  $n_j$  be the number of samples in class  $C_j$ . The class mean and global mean are defined as:

$$\boldsymbol{\mu}_j = \frac{1}{n_j} \sum_{\mathbf{x}_i \in C_j} \mathbf{x}_i, \quad \boldsymbol{\mu} = \frac{1}{n} \sum_{i=1}^n \mathbf{x}_i. \quad (1)$$

The within-class scatter  $\mathbf{S}_w$  and between-class scatter  $\mathbf{S}_b$  are:

$$\mathbf{S}_w = \frac{1}{n} \sum_{j=1}^c \sum_{\mathbf{x}_i \in C_j} (\mathbf{x}_i - \boldsymbol{\mu}_j)(\mathbf{x}_i - \boldsymbol{\mu}_j)^T, \quad (2)$$

$$\mathbf{S}_b = \frac{1}{n} \sum_{j=1}^c n_j (\boldsymbol{\mu}_j - \boldsymbol{\mu})(\boldsymbol{\mu}_j - \boldsymbol{\mu})^T. \quad (3)$$

The total scatter matrix is  $\mathbf{S}_t = \mathbf{S}_w + \mathbf{S}_b$ . An NFST projection vector  $\boldsymbol{\theta}$  must satisfy:

$$\boldsymbol{\theta}^T \mathbf{S}_w \boldsymbol{\theta} = 0, \quad \boldsymbol{\theta}^T \mathbf{S}_t \boldsymbol{\theta} > 0, \quad (4)$$

i.e. it lies in the null space of  $\mathbf{S}_w$  while staying outside the null space of  $\mathbf{S}_t$ .

### 1.2 Algorithmic Derivation

To satisfy both goals of NFST, we first address the positivity of total scatter variance ( $\boldsymbol{\theta}^T \mathbf{S}_t \boldsymbol{\theta} > 0$ ) by ensuring the projection vector  $\boldsymbol{\theta}$  lies within the range of the total scatter matrix. This is achieved by representing  $\boldsymbol{\theta}$  as a linear combination of the basis vectors of the centered data:

$$\boldsymbol{\theta} = \mathbf{Q}\boldsymbol{\alpha}. \quad (5)$$

where  $\mathbf{Q} = [\boldsymbol{\beta}_1, \dots, \boldsymbol{\beta}_{n-1}]$  is an orthonormal basis for  $\text{span}\{\mathbf{x}_i - \boldsymbol{\mu} \mid i = 1, \dots, n\}$ . By choosing  $\boldsymbol{\theta}$  from this span, we guarantee that  $\boldsymbol{\theta}^T \mathbf{S}_t \boldsymbol{\theta} > 0$  for any non-zero vector  $\boldsymbol{\alpha}$ , thus preventing the loss of discriminative information.

Next, to satisfy the vanishing of between scatter variance ( $\boldsymbol{\theta}^T \mathbf{S}_w \boldsymbol{\theta} = 0$ ), we substitute the representation  $\boldsymbol{\theta} = \mathbf{Q}\boldsymbol{\alpha}$  into the within-class scatter condition:

$$(\mathbf{Q}\boldsymbol{\alpha})^T \mathbf{S}_w (\mathbf{Q}\boldsymbol{\alpha}) = 0 \implies \boldsymbol{\alpha}^T (\mathbf{Q}^T \mathbf{S}_w \mathbf{Q}) \boldsymbol{\alpha} = 0. \quad (6)$$

This condition is equivalent to finding the null space of the projected within-class scatter matrix. We solve the following system for  $\alpha$ :

$$(\mathbf{Q}^T \mathbf{S}_w \mathbf{Q}) \alpha = 0. \quad (7)$$

Solving this system yields  $c - 1$  optimal vectors  $\alpha_i$ . The final projection vectors are recovered via  $\theta_i = \mathbf{Q} \alpha_i$ , forming the transformation matrix  $\Theta = [\theta_1, \theta_2, \dots, \theta_{c-1}]$ . Any data point  $\mathbf{x}$  is then projected as:

$$\mathbf{x}_{proj} = \Theta^T \mathbf{x}. \quad (8)$$

## 2 Kernel Mapping via Anchors

Traditional kernel-based methods, including the standard NFST, often suffer from high computational complexity, typically  $O(n^3)$ , because they require the construction and decomposition of an  $n \times n$  kernel matrix. This makes them unscalable for large-scale datasets. Furthermore, standard one-class models often assume a single unimodal distribution, failing to capture complex local manifold structures (multi-modal data). To overcome these issues, we propose an anchor-based linearization as follows.

### 2.1 Anchor-based Feature Representation

Given training samples  $\mathbf{X} = [\mathbf{x}_1, \dots, \mathbf{x}_n]^T \in \mathbb{R}^{n \times d}$ , we map them into a non-linear space using  $m$  anchors  $\mathbf{U} = [\mathbf{u}_1, \dots, \mathbf{u}_m]^T \in \mathbb{R}^{m \times d}$  ( $m \ll n$ ) obtained via **Mini-batch K-means**. For each sample  $\mathbf{x}_i$ , its similarity to anchor  $\mathbf{u}_j$  is defined by the Gaussian kernel similarity:

$$s_{ij} = \exp \left( -\frac{\|\mathbf{x}_i - \mathbf{u}_j\|^2}{\sigma} \right). \quad (9)$$

The mapped vector  $\mathbf{z}_i = [z_{i1}, \dots, z_{im}]^T \in [0, 1]^m$  is obtained by normalizing these similarities to represent the soft-membership of the sample to the anchors:

$$z_{ij} = \frac{s_{ij}}{\sum_{k=1}^m s_{ik}}. \quad (10)$$

The resulting matrix  $\mathbf{Z} = [\mathbf{z}_1, \dots, \mathbf{z}_n]^T \in \mathbb{R}^{n \times m}$  represents the data in the Kernel Anchor-Space. To capture the multi-modal structure of the one-class distribution, we further partition the training set into  $m$  sub-clusters  $\{\mathcal{C}_j\}_{j=1}^m$ . Each sample  $\mathbf{x}_i$  is assigned to a cluster via hard-partitioning based on its strongest anchor membership:

$$i \in \mathcal{C}_j \iff j = \arg \max_{k=1 \dots m} z_{ik}. \quad (11)$$

These sub-clusters  $\{\mathcal{C}_j\}$  serve as pseudo-classes for the local structural constraints and centroid calculations.

### 2.2 Dual-level Error Matrices

To capture the discriminative structure of the data, we define two error matrices representing the global and local variations in the anchor-space  $\mathbf{Z}$ :

- **Global Error ( $\Psi_G$ ):** Aiming to **shrink the entire class toward its global mean**, we use the centering matrix  $\mathbf{H} = \mathbf{I}_n - \frac{1}{n} \mathbf{1} \mathbf{1}^T$ :

$$\Psi_G = \mathbf{Z}^T \mathbf{H}. \quad (12)$$

- **Local Error ( $\Psi_L$ ):** To preserve the local manifold (subclass) structure, we use the graph Laplacian matrix  $\mathbf{L} = \mathbf{D} - \mathbf{W}$ , where  $\mathbf{W} = \mathbf{Z} \mathbf{Z}^T$  characterizes local similarities. **The local variation is captured** via:

$$\Psi_L = \mathbf{Z}^T \mathbf{L}^{1/2}. \quad (13)$$

## 2.3 Derivation of the Regularized Scatter Matrix

To find an optimal projection that balances global cohesion and local preservation, we construct a **unified regularized error matrix**  $\Psi$  by concatenating the weighted errors, controlled by  $\alpha \in [0, 1]$ :

$$\Psi = [\sqrt{\alpha}\Psi_G \quad | \quad \sqrt{1-\alpha}\Psi_L]. \quad (14)$$

The **within-class regularized scatter matrix**  $\mathbf{S}_{wreg}^\phi \in \mathbb{R}^{m \times m}$  is defined as the total variance of this unified error:

$$\begin{aligned} \mathbf{S}_{wreg}^\phi &= \Psi \Psi^T \\ &= (\sqrt{\alpha}\Psi_G) (\sqrt{\alpha}\Psi_G)^T + (\sqrt{1-\alpha}\Psi_L) (\sqrt{1-\alpha}\Psi_L)^T \\ &= \alpha(\mathbf{Z}^T \mathbf{H} \mathbf{H}^T \mathbf{Z}) + (1-\alpha)(\mathbf{Z}^T \mathbf{L}^{1/2} (\mathbf{L}^{1/2})^T \mathbf{Z}). \end{aligned} \quad (15)$$

Utilizing the property that the centering matrix  $\mathbf{H}$  is symmetric and idempotent ( $\mathbf{H}\mathbf{H}^T = \mathbf{H}$ ) and  $\mathbf{L}$  is a symmetric matrix, the expression simplifies to its final form:

$$\mathbf{S}_{wreg}^\phi = \alpha (\mathbf{Z}^T \mathbf{H} \mathbf{Z}) + (1-\alpha) (\mathbf{Z}^T \mathbf{L} \mathbf{Z}). \quad (16)$$

This matrix  $\mathbf{S}_{wreg}^\phi$  serves as the within-class scatter operator for the NFST framework, effectively integrating global compactness and local structural preservation.

## 2.4 Kernelized One-Class LPP-NFST

The Kernelized LPP-NFST finds a projection matrix  $\mathbf{V}$  in the anchor-space  $\mathbf{Z}$  that satisfies two conditions: (i)  $\mathbf{V}^T \mathbf{S}_{wreg}^\phi \mathbf{V} \rightarrow 0$  to minimize regularized within-class scatter, and (ii)  $\mathbf{V}^T \mathbf{S}_T^\phi \mathbf{V} > 0$  to preserve total variance. Following the NFST framework, the solution is derived in two stages:

1. **Range Space:** Extract an orthonormal basis  $\mathbf{Q}$  for the range of  $\mathbf{S}_T^\phi$ , ensuring Goal (ii) for any  $\mathbf{V} = \mathbf{Q}\mathbf{A}$ .
2. **Null Space:** Substitute  $\mathbf{V}$  into Goal (i) to solve the reduced eigen-system:

$$(\mathbf{Q}^T \mathbf{S}_{wreg}^\phi \mathbf{Q}) \mathbf{A} = \mathbf{0}. \quad (17)$$

The final projection  $\mathbf{V} = \mathbf{Q}\mathbf{A}$  is recovered using eigenvectors  $\mathbf{A}$  corresponding to the zero eigenvalues.

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### Algorithm 1 Kernelized One-Class LPP-NFST

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- 1: **Step 1: Mapping & Partitioning**
  - 2: - Compute  $\mathbf{Z} \in \mathbb{R}^{n \times m}$  via Eq. (10-11) using  $m$  anchors.
  - 3: - Partition samples:  $\mathcal{C}_j = \{i \mid j = \arg \max_k z_{ik}\}$ .
  - 4: **Step 2: Scatter Matrices**
  - 5: -  $\mathbf{S}_{wreg}^\phi = \alpha(\mathbf{Z}^T \mathbf{H} \mathbf{Z}) + (1-\alpha)(\mathbf{Z}^T \mathbf{L} \mathbf{Z})$  and  $\mathbf{S}_T^\phi = \mathbf{Z}^T \mathbf{H} \mathbf{Z}$ .
  - 6: **Step 3: Two-step NFST Solver**
  - 7: - Find basis  $\mathbf{Q}$  for the range of  $\mathbf{S}_T^\phi$  via SVD.
  - 8: - Solve  $(\mathbf{Q}^T \mathbf{S}_{wreg}^\phi \mathbf{Q}) \boldsymbol{\alpha}_i = \lambda_i \boldsymbol{\alpha}_i$ .
  - 9: - Set  $\mathbf{V} = \mathbf{Q}[\boldsymbol{\alpha}_1, \dots, \boldsymbol{\alpha}_p]$  where  $\lambda_i \approx 0$ .
  - 10: **Step 4: Centroids & Inference**
  - 11: - Calculate subclass centroids:  $\mathbf{b}_j = \text{mean}(\mathbf{z}_{i \in \mathcal{C}_j} \mathbf{V})$ .
  - 12: - For test  $\mathbf{z}^*$ , project  $\mathbf{y}^* = \mathbf{z}^* \mathbf{V}$ .
  - 13: - **Anomaly Score:**  $\mathcal{A}(\mathbf{x}^*) = \min_j \|\mathbf{y}^* - \mathbf{b}_j\|_2^2$ .
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